

→ Confusion Matrix (Cont.)

□ Confusion Matrix Classification

False Positive

False Negative

		Predicted class		
		Pos	Neg	
Actual class	Pos	TP	FN Type II err	Sensitivity $\frac{TP}{TP+FN}$
	Neg	Type I err	TN	Specificity
		Precision $\frac{TP}{TP+FP}$	Neg Predictive Value $\frac{TN}{TN+FN}$	Accuracy

$$F1 \text{ score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

#Scenario

- Get confusion matrix
- Get scores

02 | Shagun

SVM | Regression

Support Vector Machine (Reading)

→ R^2 is a line

→ R^3 is a plane

Regression

→ Independent variables

→ Dependent variables

→ Type

→ Linear Regression

→ Multiple Linear Regression

→ Polynomial Regression

Q. A hospital used an ML model to predict whether a patient has disease based on test results. The model was evaluated on a dataset of 500 ~~100~~ pt. among them 100 actually had the disease. The model correctly identified 80 of them as having the disease. Among the patients who didn't have a disease, the model correctly predicted 370 of them as ^{NOT} having the disease. Based on the scenario, construct confusion matrix & calculate accuracy, recall, precision & F1 score.

Shayan

Dis	TP = 80	FN = 20
No Dis	FP = 30	TN = 370
	Dis	No Dis

pred

Total dataset, $T = 500$

Diseased patients = $100 - 80 = 20$

No disease = $400 - 370$

Confusion Matrix

→ Total = Total Dataset

$$\text{Accuracy} = \frac{80 + 370}{500}$$

$$\text{Precision} = \frac{80}{80 + 30}$$

$$\text{Recall} = \frac{80}{80 + 20}$$

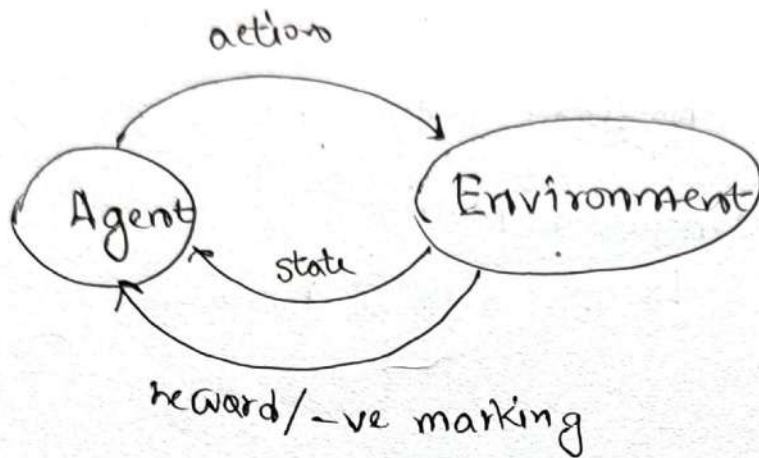
$$F_1 = 2 \times \frac{P \times R}{P + R} \times 100\%$$

Q/ Accuracy = 98%

N.S.	TP	FN
S	FP	TN
N.S	S	

"Biased"

1000 email
 ↙ 950 (not spam)
 ↘ 50 (spam)

Reinforcement Learning-

- no training data available
- no labeled data
- learning with experience (fail/pass)
- learn by taking action
- used in long term problem like game play

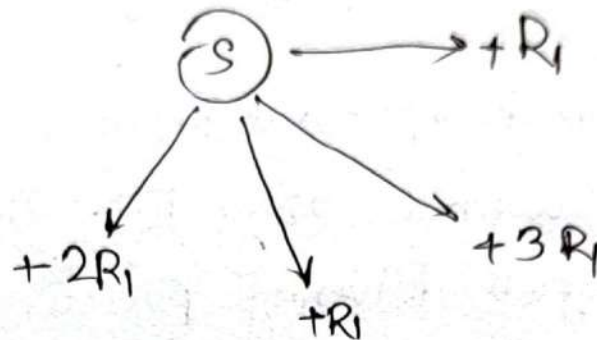
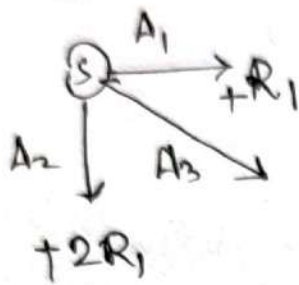
2 types.

① The reward R.L.

Where taking each action agent will be rewarded & will keep taking similar actions.

② -ve reward R.L. : less frequent actions.

① Shayan



exploitation can make an agent stuck at local maxima.

- inefficient learning
- less exploring
- stuck in local maxima

suggested strategies

- ① Modify reward fn
- ② Exploration until a new best route is found

→ Reinforcement Learning —— Problem Solving

Q. A company develops an autonomous agent to deliver packets in warehouse using R.L. The agent ^{receives} ~~gets~~ +50 for successfully delivering packets, -20 for colliding with obstacle & -1 for every movement. During training, the agent repeatedly take similar actions & rarely explore alternate patterns. Analyze how the reward fn & exploration trade-off influences agent learning. Recommend strategies to improve its navigation efficiency.

Exploration

- find more about environment
- give up immediate reward for maximizing future rewards.

Exploitation

- exploit known or profitable info about environment to maximize immediate reward